The Byzantine Dome: The Sound Dome and Wassily Kandinsky

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Abstract

This paper examines the use of sound domes for the diffusion of electronic and mixed music. These sound domes should have as their prototype, the Byzantine dome, an architectural achievement for the better diffusion of sound and light. The construction of sound domes can be proved exceptionally useful for the resolution of most of the inconveniences we encounter during the diffusion of electronic or mixed music. As an archetypal form, the dome is found in different cultures. Its use can be helpful in our endeavour to create a global sonic space culture, which can merge into it the various local characteristics and qualities.

We will attempt to focus on the application of sound domes for the transference of the composed spatial articulation into the listening environment, simultaneously examining the equivalent resultants in the form of painting as this was established by the painter Wassily Kandinsky. The theory of Kandinsky concerning form, can be useful as a guide for spatialization. The "point" – primordial element of space -, the "line" and the "plane" can be diffused through the sound dome without any previous reflections.

Introduction

A few years ago a group of musicians tried to organize a concert in the Byzantine monument of Rotonda which is situated in the city of Thessaloniki. This effort confronted the strong opposition of the Greek Orthodox Church and a part of its citizens. Finally, the concert was never performed. This example shows the intention of several musicians to put into practice the acquired knowledge of architecture and acoustics in order to perform contemporary musical works.

The construction of sound domes containing loud speakers for the diffusion of music, can be proved exceptionally useful for the resolution of most of the inconveniences we encounter during a concert of electroacoustic or mixed music. The construction of the dome as a means of tracing the utmost form of art that is, the search for the divine, commences in the distant past and ends up in the Byzantine dome; an architectural achievement for the diffusion of sound and light.

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Sound Domes

According to Kant, space and time are two "a priori" forms. This was a starting point that led Einstein to the conception of a 4-dimensional reality. It wouldn't be excessive for us to consider, that elements of this reality existed in the Byzantine architecture and art with its peak, the Byzantine dome. The conception of space was not synonymous with the one of size. Space was not formed by the means of perspective rather than, the diffusion of light. Equally, time was not the linear duration that comes

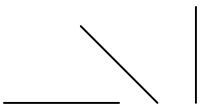
between a starting point and an end, but rather had an almost static entity, which embraces both.

Since the beginning of the 70's, Leo Kupper has been trying to complete a well-tempered sound instrument for the diffusion of electronic music through a sound dome. His installations in Grenoble in 1973, Bonn and elsewhere, led the way to a meaningful approach of spatialization. His aim has been the completion of a space modulation system with which the interpreter will be able to articulate the musical space structure. This system will function with strict and pre-determined laws on equal terms to the ones governing the pitch modulation system.

In his article "The Well Tempered Space Sound Instrument. A New Musical Instrument," published in 1991, L. Kupper considers the creation of such a system feasible. Although he mentions in detail the importance of psychological time for the perception of musical space, he is not indicating us how the psychological time will become measurable, in order to be used as one of the parameters in the space modulation system.

The Sound Dome And Wassily Kandinsky

In W. Kandinsky's treatise "Punkt und Linie zu Flache" (Point-Line-Plane) (1926), we come across many references regarding the characteristics and parameters of sound. We also encounter some elements, which even though are originally related to the form of painting, they could have been used for the spatialization of musical space. As for the painter, the primordial component of the form is the "point". W. Kandinsky considers the point as the most laconic, ephemeral form, which is characterized by excessive immobility. Then, the motion overwhelmed the point's immobility by leaping from a static to a dynamic status and thus, creating the "line". The "line" may have any direction. We will stick to the three basic archetypal types (Kandinsky 1926):

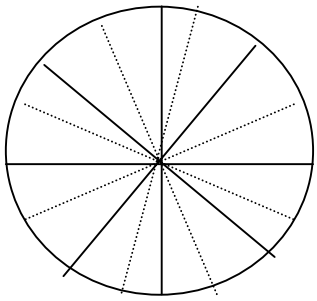


horizontal direction: cold form: potentiality of cold motion
vertical direction: hot form: potentiality of hot motion
diagonal direction: cold/hot form: potentiality of cold/hot motion

This is the point where we come across the simplest contradiction found in the form: the immobility of the point versus the motion of the line. Then, the creation of the "plane" is constructed by the combination of multi-directional lines.

These three basic components of the painting are also used in the musical form. "Point" can be considered any static percussive sound with a minimum resonance. "Line" can be considered the diffusion towards one direction of any sonic morphology, through a loud speaker. Finally, "plane" can be the multi-directional diffusion of soundscapes and images, through a combination of concentric or non-concentric loud speakers.

Another very important element of painting that we also encounter in the diffusion of musical space, is the continuum of condensation-rarefaction, which is related to the changes of temperature in the continuum of cold-hot (Kandinsky 1926).



- Concentric changes of temperature
- Concentric changes of direction

The continuum of condensation-rarefaction is qualified by the seven criteria, which are

defined by P. Schaeffer: masse, dynamique, timbre harmonique, profil melodique, profil de masse, grain, allure. Here follows an example for the diffusion of a distant space (Chion 1983):

DISTANT SPACE

→ CONDENSATION
→ HOT
→ masse : nodal group
dynamique : cyclic/resonance
timbre harmonique: complex/continuous
profil melodique : torculus
profil de masse : delta
grain : compact/discontinuous
allure : irregular

The above example is a guide for the spatialization of the composed space through a sound dome rather than, an attempt to create a space scale. The application of the three previously mentioned elements (point, line, plane) during the diffusion of the musical space, is clear and apparent. The architectural structure of the sound dome is based on the hemispherical or spherical form (two opposite-sited sound domes). The possibility of placing (geometrically or not) the loud speakers in any point of the periphery of the dome, allows the interpreter the transition from the simplest spatial component that is the point, to more complex components such as the sonic planes or soundscapes. Thus, the interpreter not only diffuses the composed space but also controls the diffused space (Smalley 1991).

Despite all these, it is exceptionally difficult, if not impossible, for a general space modulation system with specific scales and articulation structures, to be formed. The changes of temperature in the cold-hot continuum but also the combinations of condensations and rarefactions are boundless. The morphological transformations (changes in spectrum, texture) which result to spectral transformations (Smalley 1986), are not perceived by the listeners in the same way. L. Kupper admits that the perception of spectral changes and therefore the perception of space scales, is closely related to the listener's attention (Kupper 1991).

At least, the creation of a general space modulation system, is not possible without the acceptance of some very important and necessary compromises. The well-tempered pitch modulation system, on which all the horizontal and vertical structures of the instrumental music are based, neglects the microtones, the slightest timbral changes and the extremely complex rhythmical sequences which are found in the non-western music. In the same way, a well-tempered space modulation system will not contain the microscopic transitions, which are taking place inside the above continua.

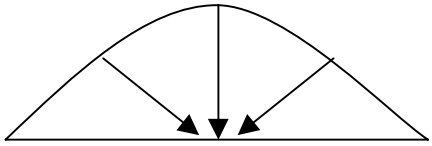
It is more essential for the interpreter to be given the freedom to use these continua as auxiliary elements for the diffusion of the musical space. The architectural structure of the sound dome offers him such a freedom. Sonic lines and directions, starting from one or more loud speakers (concentric or not) situated in different points of the dome and also, countless combinations of sonic planes and images, can all be directed towards the listener without any previous reflections. From the immobility of the "point" to the daedalian combinations of motion, the structure of the sound dome, which hovers over the audience, diffuses the sound directly to the listener. As for the mixed music, the dome with the instruments which are placed in the same level with the audience, form a sphere, a unique instrument for the spatial diffusion. This sphere embraces the listener and thus triggers off the closest connection between the spatialization and the perception of the musical space.

The sphere and therefore its half – the hemisphere – is the perfect geometrical form. It symbolizes at the same moment the infinitude and the "point" that is, the space and its primordial structural element. The sphere itself is a point consisting of other

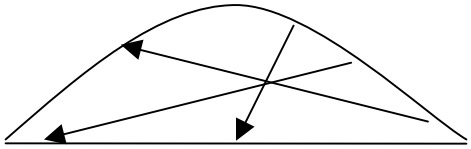
points (loudspeakers) which outline its periphery. The expansion of this periphery is boundless.

The placing of the loud speakers across the periphery of the dome can be:

1) Geometrical or concentric. In this way, the diffusion of the "lines" and the "planes" is also geometrical.
2) Non-geometrical or non-concentric. Thus, the interpreter can distract the uniformity of the concentric "lines" and "planes".



1) concentric placing



2) non-concentric placing

Conclusion

The example of the cancelled concert in the Byzantine monument of Rotonda, should not discourage us from the practise of the acquired knowledge into the contemporary musical expression.

The shape of the dome, for thousands of years now, symbolizes the utmost architectural construction for the quest of the divine. As an archetypal form, the dome is found in different cultures. Its use can be helpful in our endeavour to create a space culture. This space culture can merge into it the various local characteristics and qualities. A very important factor for the creation of this general space culture is the construction of permanent sound domes that will lead to the end of the Italian scenes or at least, their maintenance as museums.

Just like in the painting form as in the musical one, the articulation of the real and the psychological time, needs an expansion of its expressive means. We may have to stop confusing the notion of space with the one of size. We may have to stop measuring the time with rhythmical sequences. We may, finally, have to use an expressive means that will contain at the same moment the "point" and the infinitude, the diminution and the magnification, the starting point and the ending.

Using the sound dome for these purposes does not mean that we will be forced to create a general space modulation system. In any case, the pressing urge of the western civilization to decode every single evolution, should not lead us to a far-fetched syntax of such a system. By the end of this millennium there are a lot who consider typography to be the most important cultural invention. Unfortunately, this invention also encloses its degeneration since anything that is recorded ceases to evolve.

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